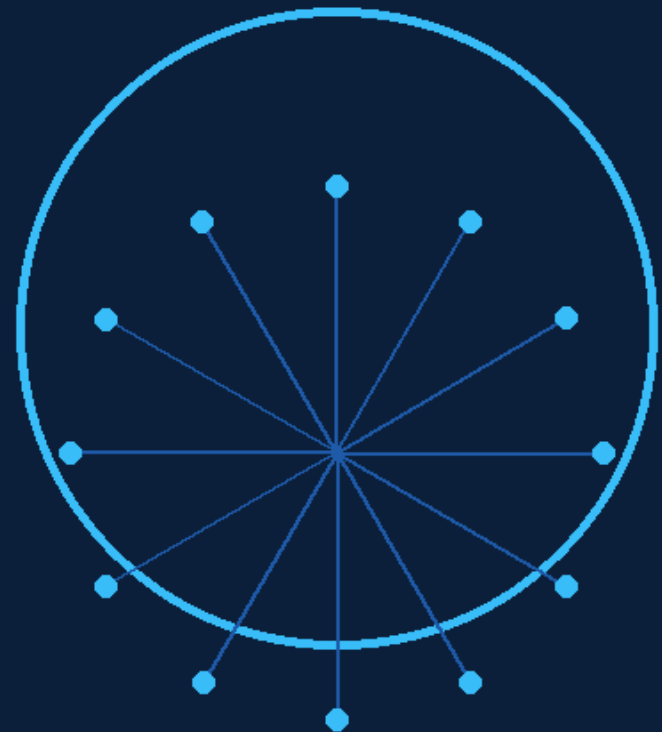


AI/ML ENGINEER INTERVIEW PLAYBOOK

Questions + Concepts you need to know

(from Head of AI/ML perspective)

- LLM Fundamentals
- Prompting & Context Engineering
- RAG Systems
- MLOps & LLMOps
- Cost & Latency Tradeoffs
- System Design Thinking
- Real-World Scenarios



★ Save this carousel & ace your next interview!

1. LLM FUNDAMENTALS



What is tokenization, and how does it affect generation?



How do embeddings really work?



What's the role of attention and positional encoding?



What changes during fine-tuning?
Optimizers, schedulers, layer freezing



LoRA vs QLoRA vs full fine-tune — what are the tradeoffs?



2. PROMPTING & CONTEXT ENGINEERING



Few-shot vs zero-shot — which works better where?



How do you design system prompts that are robust across users?



How do you make output deterministic?



How do you track, version, and backfill changing context?



How do you build and maintain memory?



3. RAG SYSTEMS



What's your chunking strategy — by length, semantics, or structure?



How do you choose a vector DB — Chroma, Pinecone, OpenSearch?



Can you update or backfill embeddings with zero downtime?



How do you evaluate retrieval quality — precision@k, reranking, citation?



4. MLOps & LLMOps



Sketch a pipeline: raw data → model → serving → feedback



How would you monitor performance drift or hallucinations?



How do you log prompts and outputs for debugging and auditing?



CI/CD for LLM workflows — what's different from ML?



DATA



MODEL



SERVING



FEEDBACK

5. COST & LATENCY TRADEOFFS



How do you reduce token usage?



When should you quantize a model?



What's your batching + caching strategy to reduce latency?



When should you use hosted APIs vs open-source models?



6. SYSTEM DESIGN THINKING



How do you make an AI system more deterministic and less brittle?



What fallback do you use if the LLM fails mid-task?



Can you solve this without an LLM or vector DB?



What's the right database — SQL, NoSQL, or vector?



7. REAL-WORLD SCENARIOS

1

What happens if your embedding model changes — how do you migrate safely?

2

How would you fine-tune a model on user behavior and deploy it?

3

How would you make this system cheaper without killing quality?

4

Can you walk me through a debugging session for incorrect LLM outputs?



**HIMANSHU
SINGHVI**

AI & Data Scientist

AI • Agentic AI • Data Science • Enterprise AI

 Building Intelligent Solutions,
Driving Real Impact.



EXTRA CONCEPTS TO KNOW



**Transformer
Architecture**

ABC

**Tokenization &
Vocabulary**



**Attention
Mechanism**



**Fine-tuning
Strategies**



**Evaluation
Metrics**



**Retrieval
Techniques**



**Guardrails
& Safety**



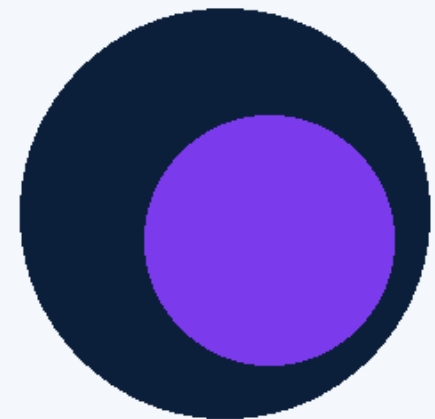
**Ethics &
Bias**

★ **Strong fundamentals + practical experience = Great AI/ML engineer**

Master the concepts. Practice the tradeoffs. Explain your thinking.

INTERVIEWER'S MINDSET

- ✓ We don't just test what you know.
- ✓ We test how you think, trade off, and solve real problems.
- ✓ Communicate your thought process clearly.
- ✓ It's okay to say "I don't know — but here's how I'd find out."
- ✓ Curiosity, ownership, and iteration > knowing everything.



Think in systems, not

QUICK TIPS TO PREPARE



Build end-to-end projects (RAG, fine-tuning, agents, etc.)



Stay updated with latest papers & tools



Practice system design for AI applications



Review common failure modes and how to fix them



Mock interviews & discuss your reasoning aloud



What's the realest
AI/ML interview question
you've encountered?



What else would
you add here?



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Let's learn together! □